

By a (pure) *automaton*, we will mean a 3-sorted algebra $\mathfrak{A} = (A, G, B)$, where A is a set of states, G a semigroup of input signals and B a set of output signals, i.e., external states, and we assume that the automaton comes with a transition operation, $\circ: A \times G \rightarrow A$, and an output operation, $\ast: A \times G \rightarrow B$, bound by the conditions $a \circ g_1 g_2 = (a \circ g_1) \circ g_2$, $a \ast g_1 g_2 = (a \circ g_1) \ast g_2$. If A and B are modules over some ring K , and the operations $\circ$ and $\ast$ are linear, then $\mathfrak{A}$ is called a *linear automaton*. An automaton $\mathfrak{A}$ is called a *Mur automaton* if there is a map $\mu: A \rightarrow B$ such that $a \ast g = (a \circ g) \mu$. It is easy to show that $\mathfrak{A}$ is a pure Mur automaton if, and only if, the quasi-identity $x_1 \circ y_1 = x_2 \circ y_2 \Rightarrow x_1 \ast y_1 = x_2 \ast y_2$ holds, and a linear automaton $\mathfrak{A}$ is a Mur automaton if, and only if, μ is a linear map, and the quasi-identity

$$\sum_{i=1}^n x_i \circ u_i = 0 \Rightarrow \sum_{i=1}^n x_i \ast u_i = 0$$

is valid, where $x_i \in A$, $y_i \in G$, and the u_i are elements of the free semigroup algebra KG^1 .

These and many other considerations make it necessary to construct a general theory of quasivarieties of automata.

Higgins [1] gave an invariant characteristic of varieties of many-sorted algebras (cf. [2] for its application to automata). The corresponding characteristic of quasivarieties, that generalizes Mal'tsev's well-known theorem [3], can be found in [4]: A class of many-sorted algebras is a quasivariety if, and only if, it is closed relative to filtered products, is hereditary and contains the identity algebra. In our paper, this characteristic is applied to automata.

Our paper consists of three sections. The first two are devoted to quasivarieties of automata. We define classes of automata, saturated relative to input and output signals. These are dual concepts.

In Sec. 1, we concern ourselves with questions about saturation relative to input signals, and in Sec. 2, saturation relative to output signals. It is shown that saturation relative to input signals is equivalent to the quasivariety being presentable by quasi-identities that do not involve any equations relating to the semigroup. On the other hand, quasivarieties of pure automata, as well as of automata linear over a field, are saturated relative to output signals if, and only if, it is possible to exclude from their defining quasi-identities all variables over the set of output signals. Moreover, if $\mathfrak{X}$ is a quasivariety of automata, saturated relative to input signals, and G is a semigroup, then we will let $\mathfrak{X}_G$ denote the fiber consisting of all automata $(A, G, B) \in \mathfrak{X}$. Each of these G -automata can be viewed as a 2-sorted algebra, the elements of G being considered as operations. The fiber is also a quasivariety. We give a determination, in terms of fibers, of quasivarieties saturated relative to input signals. This determination serves as the basis for the characteristic, mentioned above, of quasivarieties saturated with respect to input signals. We consider connections between quasivarieties of automata, saturated with respect to input signals, and quasivarieties of semigroups. To each quasivariety of automata $\mathfrak{X}$, saturated relative to input signals, there corresponds a quasivariety of semigroups $\mathfrak{X}$, consisting of all semigroups that admit a faithful representation in $\mathfrak{X}$. It is shown that this correspondence is surjective. A series of results, contained in the first two sections, generalize the corresponding results in [5].

Section 3 is devoted to an application of the methods of many-sorted algebras to the theory of quasigroups. This application is founded upon the concept of *reversible automaton*, viz., a triple (A, B, C) of sets of the same cardinality, with only one operation $\ast: A \times B \rightarrow C$, invertible on both sides. So, a reversible automaton is a 3-sorted quasigroup [6]. What we have in mind here is the reversibility of the automaton relative to the output function,

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as distinct from reversibility relative to the transition function, as considered in [7]. Every quasigroup Q can be considered as a regular reversible automaton (Q, Q, Q) . Two quasigroups Q and Q' are isotopic if, and only if, their corresponding regular reversible automata are isomorphic. This approach allows one to prove, for instance, the following propositions: If Θ is some variety (resp., quasivariety) of groups, then the class of all quasigroups, isotopic to groups in Θ , is also a variety (resp., quasivariety) (a generalization of the corresponding results in [8, 9]); every reversible automaton (A, B, C) is isomorphic to some regular automaton (Q, Q, Q) , where Q is a loop (as a consequence of this proposition, we have the well-known fact that every quasigroup is LP-isotopic to some loop [6], and the proposition reveals the real reason why this is so). We also give some other results about reversible automata.

1. QUASIVARIETIES OF AUTOMATA, SATURATED RELATIVE TO INPUT SIGNALS

Let $\mathfrak{A} = (A, G, B)$ be a semigroup automaton, S_A the semigroup of all permutations of the set A , $\text{Fun}(A, B)$ the set of all mappings from A into B and $S(A, B) = S_A \times \text{Fun}(A, B)$. By defining on $S(A, B)$ the following operation

$$(\sigma, \varphi)(\sigma', \varphi') = (\sigma\sigma', \sigma\varphi'),$$

we obtain a semigroup.

If $\mathfrak{A}$ is a linear automaton, i.e., A and B are modules over some associative, commutative ring K with identity, then we have to replace $S(A, B)$ above by the semigroup $E(A, B) = \text{End } A \times \text{Hom}(A, B)$, endowed with the analogous operation of multiplication.

The operations $\circ$ and $*$ define a homomorphism $\mu: G \rightarrow S(A, B)$ of pure automata and a homomorphism $\mu: G \rightarrow E(A, B)$ of linear automata. The kernel equivalence of this homomorphism is called the *kernel of the automaton* $\mathfrak{A}$. An automaton with trivial kernel is called *faithful*.

Unless we indicate otherwise, the term *automaton* will mean a linear semigroup automaton over some associative, commutative ring K with identity. However, the results below are also true for pure automata, as well as for group and ring automata.

Definition. A class $\mathfrak{X}$ of automata is said to be *saturated relative to inputs* if for every epimorphism $(A, G_1, B) \rightarrow (A, G_2, B)$ on inputs, the following holds: $(A, G_1, B) \in \mathfrak{X} \Leftrightarrow (A, G_2, B) \in \mathfrak{X}$.

Let (A, G, B) be a given automaton. Every element $g \in G$ defines two operations: $g^\circ: A \rightarrow A$ and $g*: A \rightarrow B$. For fixed semigroup G of input signals, the automaton can be considered as a 2-sorted system (A, B) , the elements of G being viewed as operations. The associated pair (A, B) will be called a *G-automaton*. If $\mathfrak{X}$ now denotes a class of automata and G is a semigroup, then we will let $\mathfrak{X}_G$ denote the class of G -automata (A, B) for which $(A, G, B) \in \mathfrak{X}$ (the fiber of G -automata from $\mathfrak{X}$).

THEOREM 1. If $\mathfrak{X}$ is quasivariety of automata, saturated relative to inputs, then the fiber $\mathfrak{X}_G$ is for every semigroup G a quasivariety of G -automata. Conversely, if every fiber $\mathfrak{X}_G$ of an hereditary class $\mathfrak{X}$, saturated relative to inputs, is a quasivariety, then $\mathfrak{X}$ is likewise a quasivariety.

Proof. The quasivariety $\mathfrak{X}$ contains a unit object $(o, 1, o)$, where o is the empty space, and 1 is the identity semigroup. By mapping all the elements of G to 1 , we obtain an epimorphism. Since $\mathfrak{X}$ is saturated, we get $(o, G, o) \in \mathfrak{X}$, which means that the G -automaton (o, o) belongs to $\mathfrak{X}_G$, i.e., each class $\mathfrak{X}_G$ is non-empty and contains an identity object.

If (A_1, B_1) is a G -sub-automaton of a G -automaton (A, B) in $\mathfrak{X}_G$, then from $(A, G, B) \in \mathfrak{X}$ would follow $(A_1, G, B_1) \in \mathfrak{X}$. So, by definition, $(A_1, B_1) \in \mathfrak{X}_G$, and the class $\mathfrak{X}_G$ is seen to be hereditary.

It remains to verify that the class $\mathfrak{X}_G$ is closed under filtered products. Let $(A_\alpha, B_\alpha) \in \mathfrak{X}_G$, for all $\alpha \in I$, let $\mathcal{D}$ be a filter on I , and suppose that for each $\alpha \in I$ $G_\alpha = G$. Consider the filtered product of automata $\mathfrak{A}_\alpha = (A_\alpha, G_\alpha, B_\alpha)$, and denote it by $\mathfrak{A}/\mathcal{D}$, where $\mathfrak{A} = \prod_\alpha \mathfrak{A}_\alpha$. We have $\mathfrak{A}/\mathcal{D} = (\prod_\alpha A_\alpha/\rho_1, \prod_\alpha G_\alpha/\rho_2, \prod_\alpha B_\alpha/\rho_3)$, where the congruence $\rho = (\rho_1, \rho_2, \rho_3)$ is defined by the filter $\mathcal{D}$. Since $\mathfrak{X}$ is a quasivariety, $\mathfrak{A}/\mathcal{D} \in \mathfrak{X}$. Using the epimorphism of inputs $\mathfrak{A}' = (\prod_\alpha A_\alpha/\rho_1, \prod_\alpha G_\alpha, \prod_\alpha B_\alpha/\rho_3) \rightarrow \mathfrak{A}/\mathcal{D}$, and remembering that $\mathfrak{X}$ is saturated relative to inputs, we conclude that $\mathfrak{A}' \in \mathfrak{X}$.

We will denote $\prod_{\alpha} G_{\alpha}$ by G^I . For every $g \in G$ we consider the function $\bar{g} \in G^I$, which takes on the same value g for every $\alpha \in I$: $\bar{g}(\alpha) = g$. In this way, the semigroup G is embedded in G^I as the subgroup of constant functions. As a result, the automaton $(\prod_{\alpha} A_{\alpha}/\rho_1, G, \prod_{\alpha} B_{\alpha}/\rho_3)$, being a subautomaton of an automaton $\mathfrak{U}'$ in $\mathfrak{X}$, is also in $\mathfrak{X}$. Going over now to the G -automaton $(\prod_{\alpha} A_{\alpha}/\rho_1, \prod_{\alpha} B_{\alpha}/\rho_3)$, we see that it coincides with the filtered product, relative to the filter $\mathcal{D}$, of all (A_{α}, B_{α}) . By definition, this filtered product belongs to the class $\mathfrak{X}_G$. So, $\mathfrak{X}_G$ is a quasivariety.

We will now prove the converse. Every G -fiber $\mathfrak{X}_G$ contains an identity object (o, o) . This is true, in particular, for the identity semigroup 1 , hence $(o, 1, o)$ is an identity object in $\mathfrak{X}$.

If (A_1, B_1) is a subautomaton of a G -automaton (A, B) , then $(A_1, B_1) \in \mathfrak{X}_G$ and $(A_1, G, B_1) \in \mathfrak{X}$. If H is a subsemigroup of G , then the sub-automaton (A, H, B) belongs to $\mathfrak{X}$ by hypothesis.

We proceed to verify that $\mathfrak{X}$ is closed relative to filtered products. Suppose that we are given automata $\mathfrak{U}_{\alpha} = (A_{\alpha}, G_{\alpha}, B_{\alpha}) (\alpha \in I)$ in $\mathfrak{X}$, a filter $\mathcal{D}$ on I , and a filtered product $\mathfrak{U}/\mathcal{D}$ of these $\mathfrak{U}_{\alpha}$: $\mathfrak{U}/\mathcal{D} = (\prod_{\alpha} A_{\alpha}/\rho_1, \prod_{\alpha} G_{\alpha}/\rho_2, \prod_{\alpha} B_{\alpha}/\rho_3)$. We will let $\bar{G}$ denote the Cartesian product $\prod_{\alpha} G_{\alpha}$. For every $\alpha \in I$, the projection $\bar{G} \rightarrow G_{\alpha}$ defines an automaton $(A_{\alpha}, \bar{G}, B_{\alpha})$, together with an epimorphism of inputs $(A_{\alpha}, \bar{G}, B_{\alpha}) \rightarrow (A_{\alpha}, G, B_{\alpha})$. From our hypothesis of saturation, we deduce that $(A_{\alpha}, \bar{G}, B_{\alpha}) \in \mathfrak{X}$. We now consider the $\bar{G}$ -fiber $\mathfrak{X}_{\bar{G}}$. The filtered product, relative to $\mathcal{D}$, of all the $\bar{G}$ -automata (A_{α}, B_{α}) then coincides with $(\prod_{\alpha} A_{\alpha}/\rho_1, \prod_{\alpha} B_{\alpha}/\rho_3)$, and this filtered product belongs to the fiber $\mathfrak{X}_{\bar{G}}$. But this then means that the automaton $(\prod_{\alpha} A_{\alpha}/\rho_1, \bar{G}, \prod_{\alpha} B_{\alpha}/\rho_3)$ belongs to the class $\mathfrak{X}$. We also have an epimorphism of inputs

$$(\prod_{\alpha} A_{\alpha}/\rho_1, \bar{G}, \prod_{\alpha} B_{\alpha}/\rho_3) \rightarrow (\prod_{\alpha} A_{\alpha}/\rho_1, \bar{G}/\rho_2, \prod_{\alpha} B_{\alpha}/\rho_3) = \mathfrak{U}/\mathcal{D}.$$

From the saturation hypothesis on $\mathfrak{X}$ we now conclude that $\mathfrak{U}/\mathcal{D} \in \mathfrak{X}$.

Our next aim is to show that quasivarieties of automata, saturated relative to inputs, can be defined by quasi-identities that do not involve any semigroup equations.

To this end, we proceed to define the corresponding free objects.

First, we will define what is meant by a free linear G -automaton, G being an arbitrary semigroup. Let (X, Z) be a pair of sets. Together with the semigroup G , we will also consider the semigroup G^1 , obtained from G by freely adjoining an identity element. G and G^1 span the semigroup algebras KG and KG^1 . Let $X \circ KG^1$ be a free KG^1 -module on X , $X * KG$ a free KG -module on X and KZ a free K -module on Z .

We consider the triple $W_G = (X \circ KG^1, G, X * KG \oplus KZ)$, and define on it the automaton operations $\circ$ and $*$:

$$\begin{aligned} (\sum x_i \circ u_i) \circ g &= \sum x_i \circ u_i g, \\ (\sum x_i \circ u_i) * g &= \sum x_i * u_i g \text{ for every } g \in G. \end{aligned}$$

It is easily verified that the G -automaton $(X \circ KG^1, X * KG \oplus KZ)$ obtained in this manner is a free object, generated by the sets (X, Z) , in the category of all G -automata over the ring K .

Suppose now that we are given three sets (X, Y, Z) , and let $F = F(Y)$ be the free semigroup. It is immediately seen that the automaton $W = (X \circ KF^1, F, X * KF \oplus KZ)$ is the free automaton, generated by the triple (X, Y, Z) , in the category of all linear automata over the ring K .

THEOREM 2. A quasivariety of automata $\mathfrak{X}$ can be defined by quasi-identities not involving the semigroup if, and only if, the quasivariety is saturated relative to inputs.

Proof. Sufficiency. Let $\mathfrak{X}$ be a quasivariety of automata, saturated relative to inputs, and $F = F(Y)$ a free semigroup of countable rank. By Theorem 1, $\mathfrak{X}_F$ is a quasivariety. Its

quasi-identities relate to the free F-automaton $(X \circ KF^1, X * KF \oplus KZ)$, and each is of the form

$$u_1 = o \wedge u_2 = o \wedge \dots \wedge u_n = o \Rightarrow v = o, \quad (1)$$

where the u_i and v are elements of $X \circ KF^1$ or $X * KF \oplus KZ$, and the elements of Y are considered as constants. We will show that this same collection of quasi-identities of the form (1), with variables $y_i \in Y$, define the quasivariety $\mathfrak{X}$. First, we make a general remark.

Given an automaton (A, G, B) and arbitrary maps $\mu_1: X \rightarrow A$, $\mu_2: Y \rightarrow G$, $\mu_3: Z \rightarrow B$, we can extend them to a homomorphism $\mu: W \rightarrow (A, G, B)$. We denote $F^{\mu_2} = G_0$. Then we have a sub-automaton (A, G_0, B) in (A, G, B) . By means of μ_2 , we define an automaton $(A, F, B): a \circ f = a \circ f^{\mu_2}$, $a * f = a * f^{\mu_2}$. In addition, $(A, F, B) \rightarrow (A, G_0, B)$ is an epimorphism of inputs. The map $\mu' = (\mu_1, \mu_2, \mu_3)$ of F-automata $(X \circ KF^1, X * KF \oplus KZ)$ and (A, B) is a homomorphism.

We will verify that for every element u in $X \circ KF^1$ or $X * KF \oplus KZ$, we always have $u^\mu = u^{\mu'}$.

Indeed, let $u = \sum x_i \circ u_i(y_1, \dots, y_k) \in X \circ KF^1$. Then

$$u^\mu = u^{\mu_1} = (\sum x_i \circ u_i(y_1, \dots, y_k))^{\mu_1} = \sum x_i^{\mu_1} \circ u_i(y_1^{\mu_2}, \dots, y_k^{\mu_2}) = \sum x_i^{\mu_1} \circ u_i(y_1, \dots, y_k)^{\mu_2} = \sum x_i^{\mu_1} \circ u_i(y_1, \dots, y_k) = u^{\mu'}.$$

Similarly, if $u = \sum x_i * u_i(y_1, \dots, y_k) + \sum \lambda_i z_i \in X * KF \oplus KZ$, then $u^\mu = u^{\mu_3} = \sum x_i^{\mu_1} * u_i(y_1, \dots, y_k)^{\mu_2} + \sum \lambda_i z_i^{\mu_3} = u^{\mu'}$.

We now return to our proof of sufficiency. Let $(A, G, B) \in \mathfrak{X}$. We will verify that every quasi-identity of the form (1), with variables $y_i \in Y$, is satisfied in this automaton. We will use the notation of the above remark. Together with (A, G, B) , the class $\mathfrak{X}$ also contains the subautomaton (A, G_0, B) . By the saturation property, the corresponding F-automaton (A, B) belongs to the fiber $\mathfrak{X}_F$, and the quasi-identities of the form (1), with constant y_i , therefore hold in this automaton. We now proceed from the homomorphism $\mu: W \rightarrow (A, G, B)$, and we suppose that the equations $u_1^\mu = o \wedge u_2^\mu = o \wedge \dots \wedge u_n^\mu = o$ apply, i.e., that the premises of the quasi-identities of form (1), with variables $y_i \in Y$, are satisfied in (A, G, B) . It then follows from our remark that we have at the same time

$$u_1^{\mu'} = o \wedge u_2^{\mu'} = o \wedge \dots \wedge u_n^{\mu'} = o.$$

Since the F-automaton (A, B) belongs to the class $\mathfrak{X}_F$, it follows from these last equations that $v^{\mu'} = o$. But then $v^\mu = v^{\mu'} = o$, and hence the quasi-identities of the form (1), with variables $y_i \in Y$, are valid in the class $\mathfrak{X}$.

Conversely, suppose that the quasi-identities of the form (1), with variables $y_i \in Y$, hold in (A, G, B) . We will show that $(A, G, B) \in \mathfrak{X}$. To do this for the quasivariety $\mathfrak{X}$, or indeed for any axiomatizable classes, it will be sufficient to consider the case where the automaton (A, G, B) is finitely generated. We take an epimorphism $\nu: F \rightarrow G$. Then we deduce as above that the quasi-identities of the form (1) with constant y_i hold in (A, F, B) , and hence $(A, F, B) \in \mathfrak{X}_F \subset \mathfrak{X}$. The map $(1, \nu, 1): (A, F, B) \rightarrow (A, G, B)$ is an epimorphism of inputs, and, since $\mathfrak{X}$ is saturated, we obtain $(A, G, B) \in \mathfrak{X}$.

Necessity. Suppose that the quasivariety $\mathfrak{X}$ is given by quasi-identities of the form (1), without semigroup involvement. We will prove that it is saturated relative to inputs. Suppose that we are given some epimorphism of inputs $(1, \nu, 1): (A, G, B) \rightarrow (A, H, B)$ of two automata.

Let $(A, G, B) \in \mathfrak{X}$. We use the homomorphism $\mu = (\mu_1, \mu_2, \mu_3): W \rightarrow (A, H, B)$ to construct another homomorphism $\mu_2': F \rightarrow G$ with the property $\mu_2 = \mu_2' \nu$. We do this as follows. For every $y \in Y$ we find a $g \in G$ such that $y^{\mu_2} = g$. We denote $y^{\mu_2'} = g$, and define in this way a map $\mu_2': Y \rightarrow G$, which can be uniquely extended to the desired homomorphism $\mu_2': F \rightarrow G$. It follows from the definition of μ_2' that $\mu' = (\mu_1, \mu_2', \mu_3): W \rightarrow (A, G, B)$ is a homomorphism, and we have a commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{\mu} & (A, H, B) \\ & \searrow \mu' & \uparrow (1, \nu, 1) \\ & & (A, G, B) \end{array}$$

We see that for every element u , belonging to $X \circ KF^1$ or $X * KF \oplus KZ$, the equality $u^\mu = u^{\mu'}$ applies.

Suppose now that the following premises of the implication (1) hold in (A, H, B) :

$$u_1^\mu = o \wedge u_2^\mu = o \wedge \dots \wedge u_n^\mu = o,$$

then the equations

$$u_1^{\mu'} = o \wedge u_2^{\mu'} = o \wedge \dots \wedge u_n^{\mu'} = o$$

are valid in (A, G, B) , and it follows from them that $v^{\mu'} = o$, seeing that $(A, G, B) \in \mathfrak{X}$. But, then $v^\mu = v^{\mu'} = o$ is also in (A, H, B) , whence $(A, H, B) \in \mathfrak{X}$.

Let $(A, H, B) \in \mathfrak{X}$. We use the homomorphism $\mu: W \rightarrow (A, G, B)$ to construct a homomorphism $\mu' = \mu \nu: W \rightarrow (A, H, B)$, and we see that the equation $u^\mu = u^{\mu'}$ is valid for every u belonging to $X \circ KF^1$ or $X * KF \oplus KZ$. By the same reasoning as above, we find that $(A, G, B) \in \mathfrak{X}$.

It will be convenient for what follows to introduce the following operators on classes Θ of many-sorted algebras: we denote by Θ_e the result of adjoining an identity algebra to Θ ; $S\Theta$ will denote all subalgebras of algebras in Θ ; and $C^*\Theta$ will denote all filtered products of algebras in Θ . As in the one-sorted case [3], it can be proven that the smallest quasivariety generated by the class of algebras Θ is $SC^*\Theta_e$. We will apply this to classes of automata. Moreover, for these classes we introduce yet another operator, having to do with saturation:

$V\Theta$ will be the class consisting of all automata (A, G, B) for which there exists an epimorphism of inputs $(A, G, B) \rightarrow (A, G', B)$ with right side in Θ .

It is immediately verified that if Θ is a quasivariety of automata, then $V\Theta$ is a quasivariety, saturated relative to inputs. Thus, the quasivariety of automata, saturated relative to inputs, generated by some class Θ of automata, is the class $VSC^*\Theta_e = VSC^*\Theta$.

To a given saturated quasivariety of automata, we associate a certain quasivariety of semigroups. If G is some semigroup, then we can find in each quasivariety of automata, saturated relative to inputs, an automaton (A, G, B) with given semigroup G . The question arises of the conditions under which one can find a faithful one. In connection with this question, just as for linear representations [5], we will introduce the following notation: If $\mathfrak{X}$ is a class of semigroup automata, then $\vec{\mathfrak{X}}$ will denote the class of semigroups G for which there exists a faithful automaton $(A, G, B) \in \mathfrak{X}$.

THEOREM 3. Let $\mathfrak{X}$ be a quasivariety of automata, saturated relative to inputs; then $\vec{\mathfrak{X}}$ is a quasivariety of semigroups.

Proof. $\mathfrak{X}$ contains an identity automaton $(o, 1, o)$. Therefore, the single-element semigroup belongs to the class $\vec{\mathfrak{X}}$.

Let $G \in \vec{\mathfrak{X}}$ and let H be a subsemigroup of G . Then there exists a faithful automaton $(A, G, B) \in \mathfrak{X}$. The automaton (A, H, B) , being an automaton of a faithful automaton in $\mathfrak{X}$, also belongs to $\mathfrak{X}$ and in addition is faithful. Hence, $H \in \vec{\mathfrak{X}}$, i.e., the class $\vec{\mathfrak{X}}$ is hereditary.

We proceed to prove that $\vec{\mathfrak{X}}$ is closed relative to filtered products. Let us take a system of semigroups $G_\alpha \in \vec{\mathfrak{X}}$, $\alpha \in I$, and filter $\mathcal{D}$ on I and form the filtered product $\prod_{\alpha} G_\alpha / \mathcal{D}$. For each $\alpha \in I$, we have by assumption a faithful automaton $\mathfrak{A}_\alpha = (A_\alpha, G_\alpha, B_\alpha) \in \mathfrak{X}$, with $\bar{\mathfrak{A}} = \prod_{\alpha} \mathfrak{A}_\alpha / \mathcal{D} = (\prod_{\alpha} A_\alpha / \rho_1, \prod_{\alpha} G_\alpha / \rho_2, \prod_{\alpha} B_\alpha / \rho_3)$ belonging to $\mathfrak{X}$, since $\mathfrak{X}$ is a quasivariety. The semigroup $\prod_{\alpha} G_\alpha / \rho_2$ coincides with $\prod_{\alpha} G_\alpha / \mathcal{D}$, and serves as the semigroup of input signals for the automaton $\bar{\mathfrak{A}}$ in $\mathfrak{X}$. We will check that the automaton $\bar{\mathfrak{A}}$ is faithful.

We consider, together with $\bar{\mathfrak{A}}$, the automaton $\bar{\mathfrak{A}}' = (\prod_{\alpha} A_\alpha / \rho_1, G, \prod_{\alpha} B_\alpha / \rho_3)$, where $G = \prod_{\alpha} G_\alpha$, and we introduce a special congruence τ on G :

$$g_1 \tau g_2 \Leftrightarrow I(g_1, g_2) \in \mathcal{D},$$

where $I(g_1, g_2) = \{\alpha \in I : (\forall a \in A_\alpha) a \circ g_1(\alpha) = a \circ g_2(\alpha) \wedge a * g_1(\alpha) = a * g_2(\alpha)\}$. We will verify that τ satisfies the following four conditions:

1. $\rho_2 \subset \tau$.

By definition, $g_1 \rho_2 g_2 \Leftrightarrow I_{\rho_2} = \{\alpha \in I : g_1(\alpha) = g_2(\alpha)\} \in \mathcal{D}$. Let $I_{\rho_2} \in \mathcal{D}$, then $I_{\rho_2} \subseteq I(g_1, g_2)$, since $g_1(\alpha) = g_2(\alpha) \Rightarrow a \circ g_1(\alpha) = a \circ g_2(\alpha) \wedge a * g_1(\alpha) = a * g_2(\alpha)$; hence $I(g_1, g_2) \in \mathcal{D}$, being a superset of I_{ρ_2} , and it follows that $g_1 \tau g_2$.

2. τ lies in the kernel of the automaton $\bar{\mathfrak{U}}$.

Let $g_1 \tau g_2$. We will show that for every $a \in \prod_{\alpha} A_{\alpha}$ the following equations hold: $\overline{a \circ g_1} = \overline{a \circ g_2}$ and $\overline{a * g_1} = \overline{a * g_2}$. Indeed, if $\alpha \in I(g_1, g_2)$, then $a(\alpha) \circ g_1(\alpha) = a(\alpha) \circ g_2(\alpha)$ and $a(\alpha) * g_1(\alpha) = a(\alpha) * g_2(\alpha)$. Therefore, $I(g_1, g_2)$ is a subset of the set of those $\alpha \in I$ for which the indicated equations hold. The latter set, being a superset of $I(g_1, g_2) \in \mathcal{D}$, therefore also belongs to $\mathcal{D}$, hence the following equations, $\overline{a \circ g_1} = \overline{a \circ g_2}$ and $\overline{a * g_1} = \overline{a * g_2}$, are valid. From these we deduce the desired equations that signify that τ lies in the kernel of $\bar{\mathfrak{U}}$.

3. τ is a faithful kernel.

We must check that if g_1 and g_2 are not related by τ , then they act differently via the operations $\circ$ and $*$. So, let $I(g_1, g_2) \notin \mathcal{D}$ and $\alpha \in I \setminus I(g_1, g_2)$. We can then find an a_{α} in A_{α} such that $a_{\alpha} \circ g_1(\alpha) \neq a_{\alpha} \circ g_2(\alpha)$ or $a_{\alpha} * g_1(\alpha) \neq a_{\alpha} * g_2(\alpha)$. We select an element a from $\prod_{\alpha} A_{\alpha}$ such that $a(\alpha) = a_{\alpha}$ for all $\alpha \in I \setminus I(g_1, g_2)$. We must show that for this a , $\overline{a \circ g_1} \neq \overline{a \circ g_2}$ or $\overline{a * g_1} \neq \overline{a * g_2}$. Let us assume the contrary. Then

$$\begin{aligned} (a \circ g_1)(\alpha) &= (a \circ g_2)(\alpha) = a(\alpha) \circ g_1(\alpha) = a(\alpha) \circ g_2(\alpha), \\ (a * g_1)(\alpha) &= (a * g_2)(\alpha) = a(\alpha) * g_1(\alpha) = a(\alpha) * g_2(\alpha), \end{aligned} \quad (2)$$

and all α for which the equations (2) are valid lie in $I(g_1, g_2)$, for outside this set there can be no such equations. Thus, if $\overline{a \circ g_1} = \overline{a \circ g_2}$ and $\overline{a * g_1} = \overline{a * g_2}$, then the set of all α for which the equations (2) hold belongs to $\mathcal{D}$. Its superset $I(g_1, g_2)$ must therefore also belong to $\mathcal{D}$, and it follows that $g_1 \tau g_2$. So, we have obtained a contradiction.

4. If all automata $\mathfrak{U}_{\alpha}$ are faithful, then $\tau = \rho_2$.

Let $g_1 \tau g_2$. Then $I(g_1, g_2) \in \mathcal{D}$. For $\alpha \in I(g_1, g_2)$, we have $g_1(\alpha) = g_2(\alpha)$: for every $a_{\alpha} \in A_{\alpha}$,

$$a_{\alpha} \circ g_1(\alpha) = a_{\alpha} \circ g_2(\alpha) \quad \text{and} \quad a_{\alpha} * g_1(\alpha) = a_{\alpha} * g_2(\alpha),$$

whence, by virtue of $\mathfrak{U}_{\alpha}$ being faithful, we obtain the desired result. So, the set of all $\alpha \in I$ for which $g_1(\alpha) = g_2(\alpha)$ contains the set $I(g_1, g_2)$ in $\mathcal{D}$, hence it must itself belong to $\mathcal{D}$. This, by definition, means that $g_1 \tau g_2$. Earlier, we have obtained the inclusion $\rho_2 \subset \tau$, and now we have proved its opposite, hence $\tau = \rho_2$.

Conditions 2-4 show that the automaton $\bar{\mathfrak{U}}$ is faithful, and this means that the class $\vec{\mathfrak{X}}$ is closed relative to filtered products.

It is clear from the proof that in Theorem 3 the condition of saturation relative to inputs was superfluous. However, it was inserted in the statement in order that one could connect to each semigroup at least one automaton, not necessarily faithful, that belongs to the class $\mathfrak{X}$. We will now show that to each quasivariety of semigroups $\vec{\mathfrak{Y}}$ one can associate a quasivariety of automata $\vec{\mathfrak{X}}$, saturated relative to inputs, such that $\vec{\mathfrak{X}} = \vec{\mathfrak{Y}}$.

We will consider all possible faithful automata (A, G, B) , where $G \in \vec{\mathfrak{Y}}$. We will denote the class of such automata by $\vec{\mathfrak{X}}_0$, and we will let $\omega(\vec{\mathfrak{Y}})$ denote the quasivariety of automata, saturated relative to inputs, that is generated by the class $\vec{\mathfrak{X}}_0$: $\vec{\mathfrak{X}} = \omega(\vec{\mathfrak{Y}}) = VSC^* \vec{\mathfrak{X}}_0$.

THEOREM 4. The equality $\vec{\mathfrak{X}} = \vec{\mathfrak{Y}}$ holds.

Proof. Let $G \in \vec{\mathfrak{Y}}$. We then have a faithful regular automaton (KG^1, G, KG) . This automaton, by definition, belongs to the class $\vec{\mathfrak{X}}_0$, and therefore also to the quasivariety $\vec{\mathfrak{X}}$, and, by virtue of its being faithful, we have $G \in \vec{\mathfrak{X}}$. Thus, $\vec{\mathfrak{Y}} \subseteq \vec{\mathfrak{X}}$.

Conversely, let $G \in \vec{\mathfrak{X}}$ and let (A, G, B) be a faithful automaton, belonging to the class $\vec{\mathfrak{X}} = VSC^* \vec{\mathfrak{X}}_0$. Since (A, G, B) is faithful, it belongs to the class $SC^* \vec{\mathfrak{X}}_0$, and is therefore a subautomaton of some automaton (A_1, G_1, B_1) in $C^* \vec{\mathfrak{X}}_0$. We then deduce from the definition of the operator C^* that (A_1, G_1, B_1) is a filtered product of automata from $\vec{\mathfrak{X}}_0$: $(A_1, G_1, B_1) = \prod_{\alpha} (A_{1\alpha}, G_{1\alpha}, B_{1\alpha}) / \mathcal{D}$, where $\mathcal{D}$ is a filter on I . Since $G_{1\alpha} \in \vec{\mathfrak{Y}}$ for all $\alpha \in I$, the filtered product

$G_1 = \prod_{\alpha} G_{1\alpha}/\mathcal{D}$ also belongs to the quasivariety $\mathfrak{V}$. The subsemigroup G of the semigroup G_1 likewise belongs to $\mathfrak{V}$, hence $\vec{x} \in \mathfrak{V}$.

We observe that it is possible — for instance in the case of linear automata — that there does not exist a quasivariety $\mathfrak{X}$, such that $\vec{x}$ coincides with a given quasivariety $\mathfrak{V}$ of semigroups.

2. SATURATION RELATIVE TO OUTPUT SIGNALS

Definition. A class $\mathfrak{X}$ of automata is called *saturated relative to output signals*, if the following condition holds: let (A, G, B) be some automaton and (A, G, B_0) some subautomaton with the same A and G . The class $\mathfrak{X}$ is saturated relative to output signals if always

$$(A, G, B) \in \mathfrak{X} \Leftrightarrow (A, G, B_0) \in \mathfrak{X}.$$

We will introduce yet another operator V' : $V'\mathfrak{X}$ consists of those automata (A, G, B) that admit a subautomaton (A, G, B_0) belonging to the class $\mathfrak{X}$.

One can verify that the smallest quasivariety of automata, saturated relative to output signals and generated by some class $\mathfrak{X}$, is the class $V'SC^*\mathfrak{X}$.

We will relate the condition of saturation relative to output signals to the exclusion from the quasi-identities of variables from the set Z [we remind the reader that (X, Y, Z) is a system of free generators for the free automaton W].

Proposition 1. If the variables in Z are absent from the quasi-identities that define a quasivariety $\mathfrak{X}$ of automata, then the quasivariety is saturated relative to output signals.

Proof. Suppose that $\mathfrak{X}$ is defined by quasi-identities without variables from Z , let (A, G, B) be some automaton with subautomaton (A, G, B_0) belonging to $\mathfrak{X}$. We will show that $(A, G, B) \in \mathfrak{X}$. To this end, we have to verify that every quasi-identity

$$u_1 = v_1 \wedge u_2 = v_2 \wedge \dots \wedge u_n = v_n \Rightarrow u_{n+1} = v_{n+1}$$

without variables from Z , and appearing in a presentation for the class $\mathfrak{X}$, is also valid in (A, G, B) . Suppose that we are given a homomorphism $\mu: W \rightarrow (A, G, B)$, and suppose that the following equations hold in (A, G, B) : $u_1^\mu = v_1^\mu \wedge u_2^\mu = v_2^\mu \wedge \dots \wedge u_n^\mu = v_n^\mu$. Since the variables in Z are absent from the u_i and v_i , these last equations are already valid in the subautomaton (A, G, B_0) . Since the latter lies in $\mathfrak{X}$, we also have $u_{n+1}^\mu = v_{n+1}^\mu$.

A free object in the category of pure automata is of the form $W = (X \circ F^1, F, X * F \cup Z)$, where $F = F(Y)$, and X, Y, Z are countable sets.

THEOREM 5. If $\mathfrak{X}$ is a quasivariety of pure automata, saturated relative to output signals, then it can be defined by quasi-identities without variables from Z .

Proof. We will only deal with irreducible quasi-identities, i.e., quasi-identities without repetitions by transitivity of equations.

The premise of the quasi-identity $u_1 = v_1 \wedge \dots \wedge u_n = v_n \Rightarrow u = v$ will be denoted by $\mathcal{A}$, and we will split it into three groups: the group $\mathcal{A}_1$ will be a conjunction of equations $u_i = z_i$, not containing any variables from Z , $\mathcal{A}_2$ will be the conjunction of equations of the form $u_i = z_i$, and $\mathcal{A}_3$ will be the conjunction of equations of the form $z_i = z_j$. The quasi-identity can then be written in the form

$$\mathcal{A} = \mathcal{A}_1 \wedge \mathcal{A}_2 \wedge \mathcal{A}_3 \Rightarrow u = v. \quad (3)$$

COROLLARY. $u = v$ can belong to one of the three indicated groups. We consider three cases separately.

I. $(u = v) \in \mathcal{A}_1$. We will show that in this case the condition (3) is equivalent to the quasi-identity

$$\mathcal{B} \Rightarrow u = v, \quad (4)$$

where the premise $\mathcal{B}$ is the conjunction of $\mathcal{A}_1$, as well as of the equations $u_j = v_j$, whenever either $u_j = z_j$, $v_j = z_j$ enter into the group $\mathcal{A}_2$, or $u_j = z_j$, $v_j = z_k$ enter into $\mathcal{A}_2$, with $z_j = z_k$ in $\mathcal{A}_3$.

Suppose that an automaton (A, G, B) satisfies the quasi-identity (4). We specify a homomorphism $\mu: W \rightarrow (A, G, B)$ and require that the premise of (3) be satisfied, i.e., that in

the specified notation the conjunction $\mathcal{A}^\mu = \mathcal{A}_1^\mu \wedge \mathcal{A}_2^\mu \wedge \mathcal{A}_3^\mu$ be true. But $\mathcal{B}^\mu$ is then also true. Indeed, $\mathcal{A}_1^\mu$ is more general than both $\mathcal{A}^\mu$ and $\mathcal{B}^\mu$. Furthermore, if $u_j = v_j$ appears in $\mathcal{B}$ and $\mathcal{A}_2$ contains $u_j = z_j$, $v_j = z_j$, then $u_j^\mu = z_j^\mu$, $v_j^\mu = z_j^\mu$, whence $u_j^\mu = v_j^\mu$. Finally, if $u_j = v_j$ appears in $\mathcal{B}$, where $u_j = z_j$, $v_j = z_k$ with $z_j = z_k$ in $\mathcal{A}_3$, then we have $u_j^\mu = z_j^\mu$, $v_j^\mu = z_k^\mu$, $z_j^\mu = z_k^\mu$, and hence $u_j^\mu = v_j^\mu$.

So, if $\mathcal{A}^\mu$ is true in (A, G, B) , then it follows that $\mathcal{B}^\mu$ is true, as well as its consequence $u^\mu = v^\mu$. Conversely, suppose that (3) holds in (A, G, B) . Since no variables from Z appear in (4), we can vary the definition of the homomorphism $\mu: W \rightarrow (A, G, B)$ on these variables as we please. Suppose that the premise of $\mathcal{B}^\mu$ is true in (A, G, B) , and suppose that $z_1, z_2, \dots, z_n$ constitute all the variables from Z that appear in the premise $\mathcal{A}$. We choose $z_1^\mu, z_2^\mu, \dots, z_n^\mu$ so that the premise of the quasi-identity (3) is valid.

First, it is clear that $z_1, z_2, \dots, z_n$ enter into $\mathcal{A}_2$ and $\mathcal{A}_3$. If, for some z_i ($i = 1, 2, \dots, n$) there is a v such that $v = z_i$ appears in $\mathcal{A}_2$, then we put $z_i^\mu = v^\mu$. If there is an equation $z_i = v'$ in $\mathcal{A}_2$, then the premise of (4) will contain $v = v'$, and, since the latter is true, we will have $v^\mu = v'^\mu$. Thus, the definition of z_i^μ is independent of the choice of v . If v is such that $v = z_i$ does not appear, but z_j does, with z_j satisfying the conditions $(z_i = z_j) \in \mathcal{A}_3$, $(v = z_j) \in \mathcal{A}_2$, then we put $z_i^\mu = v^\mu$. By the same argument, the definition of z_j^μ does not in this case either depend on the choice of z_j and v . For all the remaining z_i , we will assume that the z_i^μ arbitrarily coincide among one another. With such an additional specification of μ on the set Z , the premise of (3) will obviously hold, and therefore also the consequence $u^\mu = v^\mu$, which is more general than (4). So, in case I, the theorem is true, even without the assumption of $\mathfrak{E}$ being saturated.

II. $(u = v) \in \mathcal{A}$. The expression (3) takes on the form

$$\mathcal{A} = \mathcal{A}_1 \wedge \mathcal{A}_2 \wedge \mathcal{A}_3 \Rightarrow u = z. \quad (3')$$

We will consider all possible subcases here.

1) $\mathcal{A}_2$ contains the equation $w = z$.

Replacing the conclusion of (3') by $u = w$, we get the quasi-identity

$$\mathcal{A} \Rightarrow u = w, \quad (5)$$

which is equivalent to (3'). Indeed, if (3') holds in $(A, G, B) \in \mathfrak{E}$, then we have $u^\mu = z^\mu = w^\mu$ for every homomorphism $\mu: W \rightarrow (A, G, B)$, i.e., (5) also holds. Conversely, if (5) holds in (A, G, B) , then $u^\mu = w^\mu$ and $w^\mu = z^\mu$, i.e., (3') is also valid. This takes us to case I.

2) $\mathcal{A}_3$ contains an equation $z = z_i$, and $w = z$ appears in $\mathcal{A}_2$. We can convince ourselves, as above, that in this case (3) is equivalent to the quasi-identity $\mathcal{A} \Rightarrow u = w$, and this quasi-identity is without variables from Z .

3) $\mathcal{A}_3$ contains $z = z_i$, but z_i is not encountered in $\mathcal{A}_2$. By virtue of (3') being irreducible, there are no equations $z = z_j$, with $j \neq i$, in $\mathcal{A}_3$. We remove the equation $z = z_i$ from $\mathcal{A}$, and proceed to show that the resulting quasi-identity

$$\mathcal{A}' \Rightarrow u = z \quad (6)$$

is equivalent to (3').

Let (3') hold in (A, G, B) , and suppose that under some homomorphism $\mu: W \rightarrow (A, G, B)$ the premise $\mathcal{A}'$ of the quasi-identity (6) is true. By our construction, the variable z_i does not appear in $\mathcal{A}'$, hence its value z_i^μ may be chosen arbitrarily. We put $z_i^\mu = z^\mu$. The premise of (3') will then be fulfilled, hence also the conclusion $u^\mu = z^\mu$, which is more general than (6). The converse is obvious.

So, this brings us to the case where $\mathcal{A} \Rightarrow u = z$, and z does not appear in $\mathcal{A}$. We take an arbitrary homomorphism μ , under which the premise of $\mathcal{A}$ is true. The equation $u^\mu = z^\mu$ must then hold. Let b be an arbitrary element of B . Together with μ , we give ourselves a homomorphism μ' that coincides with μ on all elements except $z: z^{\mu'} = b \neq z^\mu$. Since $\mathcal{A}$ does not contain z , we can write the conditional equation $\mathcal{A}^\mu = \mathcal{A}^{\mu'}$. We also have $u^\mu = u^{\mu'}$. Since the quasi-identity $\mathcal{A} \Rightarrow u = z$ must hold under the homomorphism μ' , the truth of $\mathcal{A}^{\mu'}$ will imply the equations $u^{\mu'} = z^{\mu'} = u^\mu = b$, with the result that u^μ can equal any element of B . But this is impossible if B contains more than one element. By the saturation hypothesis, B can consist of any number of elements; so, case 3) cannot occur.

III. $(u=v) \in \mathcal{A}_3$. The expression (3) takes on the form

$$\mathcal{A} \Rightarrow z_i = z_j, \quad (3'')$$

In this solution, two subcases are possible:

1) At least one of the variables z_i, z_j appears in $\mathcal{A}_2$. We will say that the equation $v = z_j$ appears in $\mathcal{A}_2$.

(3'') can then be written in the form of a quasi-identity $\mathcal{A} \Rightarrow v = z_i$, and this takes us to case II.

2) z_i, z_j do not appear in $\mathcal{A}_2$, and $\mathcal{A}_3$ contains an equation $z_i = z_k$.

If $z_k = z_j$, then (3'') is trivially true: The conclusion is contained in the premises. We therefore have to consider the case $z_j \neq z_k$, with $z_k = v$ not appearing in $\mathcal{A}_2$, else we have the subcase 1). Then z_j can also appear in $\mathcal{A}_3$, say in the equation $(z_j = z_l) \in \mathcal{A}_3$. We will assume that this is indeed the case. Then, by removing the equation $z_j = z_l$ from $\mathcal{A}$, we obtain the quasi-identity

$$\mathcal{A}'' \Rightarrow z_i = z_j, \quad (7)$$

where $\mathcal{A}''$ is without the variable z_j . We will show that (7) is equivalent to (3'').

Suppose that (3'') holds in $(A, G, B) \in \mathcal{X}$ and that, under the homomorphism μ , the premise $\mathcal{A}''$ of the quasi-identity (7) is valid. By putting $z_l^\mu = z_j^\mu$, the premise $\mathcal{A}$ of (3''), and hence its conclusion $z_i^\mu = z_j^\mu$, become true. The latter is also the conclusion of (7). The converse is obvious.

If z_j does not appear in $\mathcal{A}''$ and does not coincide with z_i , then z_j can be any element in B , which contradicts the saturation hypothesis.

The analogous theorem also holds for linear automata over a field.

THEOREM 6. Let $\mathcal{X}$ be a quasivariety of linear automata over a field K , saturated relative to output signals, then it can be defined by quasi-identities not involving any variables from Z .

Proof. The quasi-identities under consideration may be written in the form

$$\mathcal{A}_1 \wedge \mathcal{A}_2 \Rightarrow u = v, \quad (8)$$

where $\mathcal{A}_1$ does not contain any variables from Z , but $\mathcal{A}_2$ does. We recall that the automaton $W = (X \circ KF^1, F, X * KF \oplus KZ)$ is free in $\mathcal{X}$. We consider the following possible cases:

I. $(u=v) \in \mathcal{A}_1$. We write out all the conjunctive constituents of $\mathcal{A}_2$:

$$\begin{array}{l} \sum_{i=1}^n \lambda_{1i} z_i = v_1, \\ \dots\dots\dots \\ \sum_{i=1}^n \lambda_{si} z_i = v_s, \\ \dots\dots\dots \\ \sum_{i=1}^n \lambda_{mi} z_i = v_m, \end{array}$$

where $v_k = \sum_{i=1}^r x_i * v_{ki}$, $k = 1, 2, \dots, m, x_i \in X, v_{ki} \in KF, \lambda_{ji} \in K, z_i \in Z$.

We introduce the notation

$$\sum_{i=1}^n \lambda_{ji} z_i = z'_j, \quad j = 1, 2, \dots, m.$$

Suppose that the collection of elements $z'_1, \dots, z'_s, \dots, z'_m$ from the vector space KZ contains the elements $z'_1, \dots, z'_s$ as a maximal number of linearly independent vectors.

We then have a linear system

$$\begin{cases} z'_{s+1} = \alpha_{11} z'_1 + \dots + \alpha_{1s} z'_s, \\ \dots\dots\dots \\ z'_m = \alpha_{m-s} z'_1 + \dots + \alpha_{m-s} z'_s, \end{cases}$$

which we transfer to the elements $v_1, \dots, v_s, \dots, v_m$

$$\begin{cases} v_{s+1} = \alpha_{11}v_1 + \dots + \alpha_{1s}v_s, \\ \dots \\ v_m = \alpha_{m-s-1}v_1 + \dots + \alpha_{m-s}v_s. \end{cases} \quad (9)$$

The conjunction of all the equations of (9) will be denoted by $\mathcal{A}'_2$. We will show that the quasi-identity

$$\mathcal{A}_1 \wedge \mathcal{A}'_2 \Rightarrow u = v \quad (10)$$

is equivalent to (8). Suppose that (10) holds in the automaton (A, G, B) . Suppose that we give ourselves a homomorphism $\mu: W \rightarrow (A, G, B)$ under which the premise of (8) is satisfied, i.e., the conditional expression $\mathcal{A}_1 \wedge \mathcal{A}_2^\mu$ is true. By virtue of $\mathcal{A}_2^\mu$ being true, the following equations are satisfied:

$$\begin{cases} z_1^\mu = \sum_{i=1}^n \lambda_{1i} z_i^\mu = v_1^\mu, \\ \dots \\ z_s^\mu = \sum_{i=1}^n \lambda_{si} z_i^\mu = v_s^\mu, \\ \dots \\ z_m^\mu = \sum_{i=1}^n \lambda_{mi} z_i^\mu = v_m^\mu. \end{cases} \quad (11)$$

In addition, the following equations hold

$$\begin{cases} z_{s+1}^\mu = \alpha_{11}z_1^\mu + \dots + \alpha_{1s}z_s^\mu, \\ \dots \\ z_m^\mu = \alpha_{m-s-1}z_1^\mu + \dots + \alpha_{m-s}z_s^\mu \end{cases} \quad (12)$$

and, by (11), the equations

$$\begin{cases} v_{s+1}^\mu = \alpha_{11}v_1^\mu + \dots + \alpha_{1s}v_s^\mu, \\ \dots \\ v_m^\mu = \alpha_{m-s-1}v_1^\mu + \dots + \alpha_{m-s}v_s^\mu \end{cases} \quad (13)$$

are valid as well. But then all the equations that figure in $\mathcal{A}'_2$ are valid in (A, G, B) , as well as the entire premise $\mathcal{A}_1 \wedge \mathcal{A}'_2$ of formula (10) and hence also the equation $u^\mu = v^\mu$ — a conclusion for (8).

Conversely, suppose that (8) holds in (A, G, B) . We arbitrarily extend the system of vectors $z_1, \dots, z_s$ to a basis Z' of the vector space KZ . Consider a homomorphism $\mu: W \rightarrow (A, G, B)$. Since (10) does not contain any variables from Z , it is only on the variables from X and Y that the homomorphism μ is of any interest: We can modify it as we please on Z . Suppose that the premise of (10) is true under μ , i.e., the conditional expression $\mathcal{A}_1^\mu \wedge \mathcal{A}_2^\mu$ is true. We specify a homomorphism $\nu: Z' \rightarrow B$ by the rule $z_1^{\nu} = v_1^\mu, z_2^{\nu} = v_2^\mu, \dots, z_s^{\nu} = v_s^\mu$, and we define ν in a completely arbitrary way on the remaining basis elements from Z' . We will now proceed from a new homomorphism $\mu': W \rightarrow (A, G, B)$ that only differs from μ on variables from Z , in accordance with the given rule. We will show that the conditional expression $\mathcal{A}_1^{\mu'} \wedge \mathcal{A}_2^{\mu'}$ is then true, i.e., the premise of (8) is valid. Indeed, $\mathcal{A}_1^{\mu'}$ is more general than both the premise of (8) and the premise of (10). Further, we take an arbitrary conjunctive constituent

$$v_j = \sum_{i=1}^n \lambda_{ji} z_i = z_j', \quad j = 1, 2, \dots, m$$

in $\mathcal{A}_2$, and we proceed to show that the equation

$$v_j^{\mu'} = \sum_{i=1}^n \lambda_{ji} z_i^{\mu'}$$

applies. We consider two cases.

1) $j \leq s$. In this case, by the definition of u' , $z_j^{\mu'}$ coincides with $v_j^{\mu'}$. Since μ' is a homomorphism,

$$z_j^{\mu'} = v_j^{\mu'} = \sum_{i=1}^n \lambda_{ji} z_i^{\mu'}.$$

2) $s < j \leq m$. Let $j = s + k$. We then have the relation

$$z'_j = \alpha_{k1} z'_1 + \dots + \alpha_{ks} z'_s.$$

At the same time, the conjunctive constituent $v_j = \alpha_{k1} v_1 + \dots + \alpha_{ks} v_s$ appears in $\mathcal{A}'_2$, and hence the equation

$$v_j^\mu = \alpha_{k1} v_1^\mu + \dots + \alpha_{ks} v_s^\mu$$

holds. No variables from Z appear in the last equation, hence μ can be replaced here by μ' :

$$v_j^{\mu'} = \alpha_{k1} v_1^{\mu'} + \dots + \alpha_{ks} v_s^{\mu'} = \alpha_{k1} z_1^{\mu'} + \dots + \alpha_{ks} z_s^{\mu'}.$$

Since

$$z'_j = \sum_{i=1}^n \lambda_{ji} z_i = \alpha_{k1} z'_1 + \dots + \alpha_{ks} z'_s,$$

we can apply μ' to both sides, and obtain the desired equation.

So, all the equations are true in $\mathcal{A}_2^{\mu'}$, and the premise $\mathcal{A}_1^\mu \wedge \mathcal{A}_2^\mu$ is valid. Therefore, $u^\mu = v^\mu$, from which $u^\mu = v^\mu$, since u, v contain no variables from Z . Thus, (10) holds in (A, G, B) .

II. $(u=v) \in \mathcal{A}_2$. In this case, we can write (8) in the form

$$\mathcal{A} = \mathcal{A}_1 \wedge \mathcal{A}_2 \Rightarrow u = \sum_{i=1}^n \lambda_i z_i = z'. \quad (14)$$

We will consider the following two possibilities:

1) z' belongs to the linear hull of the vectors $z'_1, \dots, z'_s$. Then $z' = \alpha_1 z'_1 + \dots + \alpha_s z'_s$. One can prove, as in Theorem 5, that the quasi-identity $\mathcal{A} \Rightarrow u = \alpha_1 v_1 + \dots + \alpha_s v_s$ is equivalent to (14), and this takes us to case I.

2) z' does not belong to the linear hull of the vectors $z'_1, \dots, z'_s$. Let $\mu: W \rightarrow (A, G, B)$ be an arbitrary homomorphism under which the premise of (14) is valid. We then have, in particular, the equations

$$v_j^\mu = z_j^{\mu'}. \quad (15)$$

The conclusion $u^\mu = z^{\mu'}$ must hold as well. Since z' does not appear in the linear hull of the vectors $z_1, \dots, z_s, \dots, z_m$, we can modify μ on z' : we define a homomorphism μ' that coincides with μ on the variables from $X \cup Y \cup \{z'_1, \dots, z'_m\}$, but differs from μ on the variable z' . For $z^{\mu'}$ we can take the zero o of the space B . In addition, μ' may, generally speaking, also differ from μ on the variables $z_1, \dots, z_n$. Nevertheless, the conditional expression $\mathcal{A}_1^{\mu'}$ coincides with $\mathcal{A}_1^\mu$, and $\mathcal{A}_2^{\mu'}$ with $\mathcal{A}_2^\mu$, since (15) still holds when μ is replaced by μ' . Hence, the premise of (14) and the conclusion $u^{\mu'} = z^{\mu'} = u^\mu = o$ hold under the homomorphism μ' . So, we obtain that u^μ equals zero and also equals any other element of B , whence $B = 0$. But, this is impossible by the saturation hypothesis; therefore, case 2) does not occur.

The above procedure does not work if, instead of a field K , we proceed from a ring K . The following arguments will then apply. Let B be a module over the ring K , and suppose that it contains two distinct elements b_1 and b_2 such that $\lambda b_1 = \lambda b_2$ for some $\lambda \in K$. We consider the quasi-identity

$$\mathcal{A} = (u = z'_1 \wedge z_1 = \lambda_1 z_2 \wedge \lambda_2 z_1 = z) \Rightarrow v = z_2, \quad (16)$$

where u, v do not contain any variables from Z , and the element $\lambda_1 \lambda_2$ is distinct from the identity element of the ring K . By applying the procedure that we used to prove Theorem 6 (case II), we get the quasi-identities

$$\begin{cases} \mathcal{A} \Rightarrow \lambda_1 v = u, \\ \mathcal{A} \Rightarrow \lambda_2 u = v. \end{cases} \quad (17)$$

Thus, both quasi-identities (17) follow from (16). However, the converse is false in general. Indeed, one can deduce from the equations $\lambda_1 v = u$, $\lambda_2 u = v$ and the premise $\mathcal{A}$ that $\lambda_1 v = \lambda_1 \lambda_2 z_2$ and $\lambda_1 \lambda_2 v = z_2$; however it does not follow from this, under our current assumptions, that $v = z_2$.

On the other hand, (16) defines a saturated quasivariety $\mathcal{K}$. Indeed, let (A, G, B_0) be a subautomaton of an automaton (A, G, B) that belongs to $\mathcal{K}$. We will show that $(A, G, B) \in \mathcal{K}$.

To this end, we consider a homomorphism $\mu: W \rightarrow (A, G, B)$, under which the premise $\mathcal{A}$ is true. Then $u^u \in B_0$, since

$$u = \sum_{i=1}^m x_i * u_i, \quad u_i \in KF,$$

hence also $z_1^u \in B_0$. Thus, the premise $\mathcal{A}$ already holds in (A, G, B_0) , and the conclusion of (16) is therefore also valid in this automaton. So, we have established that (16) holds in (A, G, B) .

We now come to the following general problem: In which cases can a quasivariety of automata $\mathcal{K}$ be defined by a collection of quasi-identities that splits into two groups, with one containing quasi-identities that do not admit any variables from Z , and the other consisting of quasi-identities that are written entirely in variables from Z ?

In essence, the second group will determine quasivarieties of modules of output signals, whereas the first would only define quasivarieties of automata that are saturated relative to output signals.

We saw, in the case of a field, that every quasivariety of automata, saturated relative to output signals, has this property.

We observe that the concepts relating to automata being saturated relative to input and output signals can be carried over to arbitrary many-sorted algebras.

3. REVERSIBLE AUTOMATA

In this section, we will consider automata (A, B, C) with a single operation of outputs $*$: $A \times B \rightarrow C$, where A , B , and C are, respectively, the sets of states, of input signals and of output signals. For these automata, the concepts of *subautomaton*, *homomorphism*, *Cartesian product*, etc., are defined in the usual manner. The indicated automaton is said to be *reversible* if, in the equation $a * b = c$ every pair of elements uniquely determines the third, i.e., knowing an output signal and a state, the given input signal is uniquely determined; also, knowing the input and output signals, the state can be uniquely established.

It is obvious that the Cartesian product of reversible automata is again a reversible automaton. However, a subautomaton of a reversible automaton need not be reversible; for instance, the triple (A, b, C) , where $b \in B$, is a subautomaton of the reversible automaton (A, B, C) , but it is not reversible. It is easily seen that in a reversible automaton the sets A , B , and C must be of equal cardinality. The homomorphic image of a reversible automaton is not necessarily reversible. For instance, let (A, B, C) be a reversible automaton and $\mu_3: C \rightarrow C'$ any map. We define an automaton (A, B, C') by putting $a * b = (a * b)^{\mu_3}$, where the $*$ in brackets refers to the original reversible automaton. The resulting automaton (A, B, C') is the homomorphic image of (A, B, C) . The set C' can be taken to consist of a single element, and (A, B, C') will then not always be reversible.

It is appropriate, in view of the above, to consider a reversible automaton as a structure with three operations, viz., the operation $*$ and its two inverses $/$ and $\backslash$, where $a * b = c \Leftrightarrow a \backslash c = b \Leftrightarrow c / b = a$. In addition, the following identities hold for arbitrary elements $a \in A$, $b \in B$, and $c \in C$:

$$\begin{cases} a * (a \backslash c) = c; & (c / b) * b = c, \\ a \backslash (a * b) = b, & (a * b) / b = a. \end{cases}$$

If the three sets A , B , and C coincide, then this defines at the same time the structure of a quasigroup. A reversible automaton is a quasigroup split into three parts, i.e., a 3-sorted quasigroup.

If, on the other hand, Q is some quasigroup, then there is a corresponding regular reversible automaton (Q, Q, Q) . In addition, a homotopy (resp., isotopy) of quasigroups $Q \rightarrow Q'$ is nothing else than a homomorphism (resp., isomorphism) of the corresponding regular automata.

The theory of reversible automata is close to the theory of quasigroups, however it is important to make the distinction. If, for instance, a and b are arbitrary elements in A and B , respectively, and $c = a * b$, then the triple (a, b, c) is a reversible subautomaton in (A, B, C) , with singleton underlying sets. Every such subautomaton of a reversible automaton will be called an *identity reversible automaton*.

The concept of reversible automaton has applications, not only in the theory of automata, but also in the theory of quasigroups. We will consider some of these applications in this section.

Let $Q(*)$ be an arbitrary quasi-group, A, B , and C three of its subsets, having the following property: if we have an equation $a * b = c$, then

- 1) $a \in A$ and $b \in B$ imply $c \in C$,
- 2) $a \in A$ and $c \in C$ imply $b \in B$,
- 3) $b \in B$ and $c \in C$ imply $a \in A$.

In this case, the triple (A, B, C) can be considered as a reversible automaton. Such an automaton will be called an *automaton in the quasi-group $Q(*)$* .

If $\mu = (\alpha, \beta, \gamma)$ is a homotopy of the quasigroup Q into a quasigroup Q' , preserving all three quasigroup operations, then $(Q^\alpha, Q^\beta, Q^\gamma)$ is obviously an automaton in the quasigroup Q' .

One can consider, in particular, automata in groups. Let G be a group, H a subgroup, and $x, y \in G$. Then the triple $(Hx, x^{-1}Hxy, Hxy)$ is an automaton in the group G :

$$\begin{aligned} h_1 x \cdot x^{-1} h_2 xy &= h_1 h_2 xy \in Hxy, \\ \text{if } a \cdot x^{-1} h_2 xy &= h_3 xy, \text{ then } a = h_3 h_2^{-1} x \in Hx, \\ \text{if } h_1 x \cdot b &= h_3 xy, \text{ then } b = x^{-1} h_1^{-1} h_3 xy \in x^{-1} Hxy. \end{aligned}$$

Moreover, every such automaton is isomorphic to the regular automaton (H, H, H) , the isomorphism $\mu = (\mu_1, \mu_2, \mu_3)$ being defined by $h^{\mu_1} = hx$, $h^{\mu_2} = x^{-1}hxy$, $h^{\mu_3} = hxy$: if $h_1 h_2 = h_3$ then

$$\begin{aligned} (h_1 h_2)^{\mu_3} &= h_1 h_2 xy = h_1^{\mu_1} h_2^{\mu_2}, \\ (h_1^{-1} h_3)^{\mu_2} &= x^{-1} h_2 xy = (h_1^{\mu_1})^{-1} \cdot h_3^{\mu_3}, \\ (h_3 h_2^{-1})^{\mu_1} &= h_1 x = h_3^{\mu_3} \cdot (h_2^{\mu_2})^{-1}. \end{aligned}$$

THEOREM 7. If (A, B, C) is an automaton in a group $G(\cdot)$, then there exist a subgroup $H \subset G$ and elements $x, y \in G$ such that $A = Hx$, $B = x^{-1}Hxy$, $C = Hxy$ and the automaton coincides with the automaton defined by the subgroup H and the elements x, y .

Proof. First, we will show that if all three sets A, B , and C contain the identity element $e \in G$, then they all coincide with each other, i.e., with some subgroup of G .

Indeed, $b = eb \in C$, i.e., $B \subseteq C$, $c = e \cdot c = e^{-1}c \in B$, i.e., $C \subseteq B$, and hence $B = C$. One proves similarly that $A = C$.

We will now fix elements $a \in A$, $b \in B$, and $c = ab \in C$. The map $\mu = (\mu_1, \mu_2, \mu_3) : (A, B, C) \Rightarrow (Aa^{-1}, aBb^{-1}a^{-1}, Cb^{-1}a^{-1})$ defined by $a^{\mu_1} = a'a^{-1}$, $b^{\mu_2} = ab'b^{-1}a^{-1}$, $c^{\mu_3} = c'b^{-1}a^{-1}$ for every $a' \in A$, $b' \in B$, and $c' \in C$ is, obviously, an isomorphism, and hence $(Aa^{-1}, aBb^{-1}a^{-1}, Cb^{-1}a^{-1})$ is a reversible automaton in G . But, since $a \in A$, $b \in B$, and $c \in C$, it follows that $e \in Aa^{-1} \cap aBb^{-1}a^{-1} \cap Cb^{-1}a^{-1}$, and we have shown that these sets coincide with some set H , where $H(\cdot)$ is a subgroup of $G(\cdot)$. Thus $A = Ha$, $B = a^{-1}Hab$, $C = Hab$.

We have shown above that the automaton $(Ha, a^{-1}Hab, Hab)$ is isomorphic to the regular automaton (H, H, H) , where H is a subgroup of G . So, we may say, up to isomorphism, that all automata in groups are regular.

Besides the concept of an automaton in a quasigroup, we will also consider quotient automata of quasigroups. Let $Q(*)$ be a quasigroup and ρ_1, ρ_2, ρ_3 three equivalence relations on the set Q , such that, given two equations $a * b = c$ and $a' * b' = c'$, we have

- 1) $a\rho_1 a' \wedge b\rho_2 b' \Rightarrow c\rho_3 c'$,
- 2) $a\rho_1 a' \wedge c\rho_3 c' \Rightarrow b\rho_2 b'$,
- 3) $b\rho_2 b' \wedge c\rho_3 c' \Rightarrow a\rho_1 a'$.

Such a triple of equivalence relations will be called an *automaton equivalence in the quasigroup*.

Under these conditions, the triple of sets $(Q/\rho_1, Q/\rho_2, Q/\rho_3)$ define a reversible automaton, which is a quotient of $Q(*)$. The basic operation of this automaton is defined by the equation

$$[a]_{\rho_1} * [b]_{\rho_2} = [a * b]_{\rho_3}.$$

The inverses of this operation are also defined in the corresponding manner.

Proposition 2. If $\rho = (\rho_1, \rho_2, \rho_3)$ is an automaton equivalence in a loop $Q(*)$, then $\rho_1 = \rho_2 = \rho_3 = \rho$ is a loop congruence of $Q(*)$.

Proof. Let $b\rho_2b'$. Then $b = e b \rho_3 e b' = b'$, where e is the identity element of $Q(*)$, and hence $\rho_2 \subset \rho_3$. Conversely, if $c\rho_3c'$ then the equations $c = ec$, $c' = ec'$ imply $c\rho_2c'$, i.e., $\rho_3 \subset \rho_2$, and hence $\rho_2 = \rho_3$. One can verify in an analogous manner that $\rho_1 = \rho_3$, and it then follows from the definition of an automaton equivalence that ρ is a loop congruence of $Q(*)$.

From Proposition 2 follows, in particular, that if G is a group, then each of its quotient automata is a regular automaton of the form $(G/H, G/H, G/H)$, where H is a normal subgroup of G .

We will apply the above concepts to prove the following theorem.

THEOREM 8. Let Θ be some variety of groups and let $\bar{\Theta}$ denote the class of quasigroups isotopic to groups in Θ . Then $\bar{\Theta}$ is a variety of quasigroups.

Proof. Let $Q \in \bar{\Theta}$, and let Q_1 be a subquasigroup of Q . We will show that $Q_1 \in \bar{\Theta}$. By hypothesis, there is an isotopy $\mu = (\alpha, \beta, \gamma)$ of a quasigroup Q with a group $G \in \Theta$. We consider the three sets $(Q_1^\alpha, Q_1^\beta, Q_1^\gamma)$ in G . Since Q_1 is a subquasigroup and μ defines an isomorphism of the regular automaton (Q_1, Q_1, Q_1) with $(Q_1^\alpha, Q_1^\beta, Q_1^\gamma)$, it follows that the latter triple is an automaton in the group G . By the above, the latter is isomorphic to some regular automaton (H, H, H) , where H is a subgroup of G . Hence, the automata (Q_1, Q_1, Q_1) and (H, H, H) are isomorphic, and the quasigroup Q_1 is therefore isotopic to the group H . But, since $H \in \Theta$, we have $Q_1 \in \bar{\Theta}$.

If ρ is a congruence on the quasigroup $Q \in \bar{\Theta}$, then the quotient quasigroup Q/ρ belongs to $\bar{\Theta}$. Indeed, we have by assumption an isotopy $\mu = (\alpha, \beta, \gamma): Q \rightarrow G \in \Theta$. Hence, we have an isomorphism of automata (Q, Q, Q) and (G, G, G) . We will consider the corresponding images of the congruence ρ , i.e., $(\rho^\alpha, \rho^\beta, \rho^\gamma)$. This is an automaton equivalence in the group G , and we have an isomorphism of the automata $(Q/\rho, Q/\rho, Q/\rho)$ and $(G/\rho^\alpha, G/\rho^\beta, G/\rho^\gamma)$. As we already observed, ρ^α, ρ^β , and ρ^γ coincide among each other, and define a normal subgroup H of G . The automaton $(Q/\rho, Q/\rho, Q/\rho)$ is then isomorphic to the regular automaton $(G/H, G/H, G/H)$, from which it follows that the quasigroup Q/ρ is isotopic to the group $G/H \in \Theta$, and, consequently, $Q/\rho \in \bar{\Theta}$.

We now take a collection of quasigroups $Q_i, i \in I$, belonging to the class $\bar{\Theta}$. For each $i \in I$ there is a group $G_i \in \Theta$ such that one has an isotopy $\mu_i = (\alpha_i, \beta_i, \gamma_i): Q_i \rightarrow G_i$. Obviously, all the isotopies $\mu_i, i \in I$, define an isotopy $\mu = \{\mu_i, i \in I\}: \prod_{i \in I} Q_i \rightarrow \prod_{i \in I} G_i$ of Cartesian products, and hence $\prod_{i \in I} Q_i \in \bar{\Theta}$.

A class of quasigroups that is closed under arbitrary isotopies will be called *universal*. Theorem 8 gives us a series of universal varieties of quasigroups.

Given any class $\mathfrak{X}$ of quasigroups, we let $\tilde{\mathfrak{X}}$ denote the class of all groups in $\mathfrak{X}$. It is easily seen that if $\mathfrak{X}$ is a variety (resp., quasivariety), then $\tilde{\mathfrak{X}}$ is a variety (resp., quasivariety).

THEOREM 9. If Θ is a variety of groups, and $\bar{\Theta}$ is the corresponding variety of quasigroups, then $\tilde{\bar{\Theta}} = \Theta$.

Proof. Let $G \in \tilde{\bar{\Theta}}$. Then $G \in \bar{\Theta}$, which means that there exists a group $G' \in \Theta$ to which G is isotopic, and hence is isomorphic, by Albert's theorem [5]. So $\tilde{\bar{\Theta}} \subset \Theta$. On the other hand, every group in Θ is isotopic (resp., isomorphic) to itself, so that, by definition, $\Theta \subset \tilde{\bar{\Theta}}$. But, since $\tilde{\bar{\Theta}}$ consists of all those groups that belong to $\bar{\Theta}$, it follows that $\Theta \subset \tilde{\bar{\Theta}}$.

THEOREM 10. Let Θ be a quasivariety of groups, and $\bar{\Theta}$ the class of quasigroups isotopic to groups $G \in \Theta$. Then $\bar{\Theta}$ is a quasivariety of quasigroups.

Proof. In accordance with Theorem 8, the class $\bar{\Theta}$ is hereditary. The identity quasigroup is obviously isotopic to the identity group, hence belongs to $\bar{\Theta}$. It then remains to observe that isotopy carries over to filtered products.

In the case where Θ is the class of all groups, the quasivariety of quasigroups $\bar{\Theta}$ can be given by the quasi-identity

$$x_1 y_2 = x_2 y_1 \wedge x_1 y_1 = x_3 y_2 \wedge x_1 y_4 = x_2 y_3 \Rightarrow x_3 y_4 = x_4 y_3,$$

called the *Reidemeister closure condition* [6]. For varieties Θ of groups, we also have the relation $\bar{\bar{\Theta}} = \Theta$.

We now go over to the analogous results for reversible automata. Let Θ be a class of groups. We will let $\bar{\Theta}$ denote the class of reversible automata (A, B, C) , isomorphic to the regular automaton (G, G, G) , where $G \in \Theta$.

THEOREM 11. 1) If Θ is a variety of groups, then $\bar{\Theta}$ is a variety of automata.

2) If Θ is a quasivariety of groups, then $\bar{\Theta}$ is a quasivariety of automata.

Proof. We will first prove 1. Let $(A, B, C) \in \bar{\Theta}$ and let (A', B', C') be a reversible subautomaton of (A, B, C) . By hypothesis, there exists a group $G \in \Theta$ such that we have an isomorphism $\mu = (\mu_1, \mu_2, \mu_3): (A, B, C) \rightarrow (G, G, G)$. As before, we will consider the triple $(A'^{\mu_1}, B'^{\mu_2}, C'^{\mu_3})$. It is an automaton in G , isomorphic to the regular automaton (H, H, H) , where H is a subgroup of G . Hence, the automaton (A', B', C') is also isomorphic to a regular automaton (H, H, H) , where $H \in \Theta$. Therefore, $(A', B', C') \in \bar{\Theta}$ and the class $\bar{\Theta}$ is hereditary.

We will now suppose that (A, B, C) in $\bar{\Theta}$ admits a quotient automaton $(A/\rho_1, B/\rho_2, C/\rho_3)$, and prove that the latter also belongs to $\bar{\Theta}$. The isomorphism of automata $\mu: (A, B, C) \rightarrow (G, G, G)$ induces an isomorphism of quotient automata. The triple $(\rho_1^{\mu_1}, \rho_2^{\mu_2}, \rho_3^{\mu_3})$ is an automaton equivalence in G , since we have an isomorphism of automata. Hence $\rho_1^{\mu_1} = \rho_2^{\mu_2} = \rho_3^{\mu_3} = \rho$, where ρ is a congruence in the group G , defined by some normal subgroup H . But then we conclude that the automaton $(A/\rho_1, B/\rho_2, C/\rho_3)$ is isomorphic to some regular automaton $(G/H, G/H, G/H)$, where $G/H \in \Theta$, and this shows that the class $\bar{\Theta}$ is closed relative to homomorphic images.

It follows from the definition that the class Θ is closed relative to Cartesian products. We will now prove 2. Let Θ be a quasivariety of groups. We have already verified that the class $\bar{\Theta}$ is hereditary.

Every reversible automaton of the form (a, b, ab) is isomorphic to the regular automaton (e, e, e) , where $e \in \Theta$, hence $(a, b, ab) \in \bar{\Theta}$.

Let us consider reversible automata (A_i, B_i, C_i) , $i \in I$, in the class $\bar{\Theta}$. We will let $\mu_i = (\alpha_i, \beta_i, \gamma_i)$ denote an isomorphism between (A_i, B_i, C_i) and a grouplike regular automaton (G_i, G_i, G_i) , where $G_i \in \Theta$, $i \in I$. Let $\mathcal{D}$ be a filter on the set I , and let (A, B, C) be the filtered product, relative to $\mathcal{D}$, of all the (A_i, B_i, C_i) . We will consider at the same time the filtered product, relative to the same filter $\mathcal{D}$, of all the groups G_i , and we will denote it by G . We will show that the automaton (A, B, C) is isomorphic to (G, G, G) . First of all, we observe that the Cartesian product of the regular automata (G_i, G_i, G_i) coincides with the regular automaton (G', G', G') , where G' is the Cartesian product of all the G_i . Suppose, moreover, that (A', B', C') is the Cartesian product of all the (A_i, B_i, C_i) . We will define a bijective map $\mu = (\alpha, \beta, \gamma): (A', B', C') \rightarrow (G', G', G')$ as follows: for all $a' \in A'$, $b' \in B'$, $c' \in C'$, we put $a'^{\alpha}(i) = [a'(i)]^{\alpha_i}$, $b'^{\beta}(i) = [b'(i)]^{\beta_i}$, $c'^{\gamma}(i) = [c'(i)]^{\gamma_i}$. Then, if $a' * b' = c'$ and is the operation of G , the following equations $\otimes$ must hold

$$\begin{aligned} (a' * b')^{\gamma} &= a'^{\alpha} \otimes b'^{\beta}, \\ (a' \setminus c')^{\beta} &= (a'^{\alpha})^{-1} \otimes c'^{\gamma}, \\ (c' / b')^{\alpha} &= c'^{\gamma} \otimes (b'^{\beta})^{-1}, \end{aligned}$$

where x^{-1} denotes the element that is the inverse of x in the group G' . It follows from this that μ is an isomorphism.

Let $\rho = (\rho_1, \rho_2, \rho_3)$ be the kernel congruence of the automaton (A', B', C') defined by the filter $\mathcal{D}$, and let μ be the congruence defined in G by the same filter. We will show that μ induces an isomorphism $\bar{\mu} = (\bar{\alpha}, \bar{\beta}, \bar{\gamma})$ of automata $(A'/\rho_1, B'/\rho_2, C'/\rho_3)$ and $(G/\tau, G/\tau, G/\tau)$. It is enough, for this purpose, to verify that $a_1 \rho_1 a_2'$ applies if, and only if, $a_1^{\alpha} \tau a_2'^{\alpha}$ applies, and similarly for ρ_2 and ρ_3 . By definition, $a_1 \rho_1 a_2'$ applies if, and only if, the set

$I_0 = \{i \in I : a'_1(i) = a'_2(i)\}$ belongs to $\mathcal{D}$. Let $I_0 \in \mathcal{D}$ be given. We will prove that $\alpha_1'^\alpha \tau \alpha_2'^\alpha$. We compute $\alpha_1'^\alpha$ and $\alpha_2'^\alpha$ on each $i \in I$. By definition, $a'_1(i) = [a'_1(i)]^{\alpha_i} = [a'_2(i)]^{\alpha_i} = a'_2(i)$. This means that $\alpha_1'^\alpha$ and $\alpha_2'^\alpha$ coincides on each $i \in I_0$, hence $\alpha_1'^\alpha \tau \alpha_2'^\alpha$, since $I_0 \in \mathcal{D}$. The converse can be verified in the same way: from $\alpha_1'^\alpha \tau \alpha_2'^\alpha$ follows $\alpha_1'^{\rho_1} \alpha_2'^{\rho_1}$. Similar arguments carry through for ρ_2 and ρ_3 . So, the class $\bar{\Theta}$ is closed relative to filtered products.

Remarks. 1) The above results show that the variety of all reversible automata contains quite a lot of subvarieties and subquasivarieties.

2) It is easily seen that if two varieties (resp., quasivarieties) of groups Θ_1 and Θ_2 are distinct, then the varieties (resp., quasivarieties) $\bar{\Theta}_1$ and $\bar{\Theta}_2$ of automata are likewise distinct. Indeed, let G be a group belonging to Θ_1 but not to Θ_2 . Then the regular automaton (G, G, G) belongs to $\bar{\Theta}_1$. Suppose that (G, G, G) also belongs to $\bar{\Theta}_2$. By hypothesis, we can then find a group H in Θ_2 , such that the regular automata (G, G, G) and (H, H, H) are isomorphic. This means that the groups G and H are isotopic, but it is well known that G and H must then also be isomorphic. So, G lies in Θ_2 , which contradicts the hypothesis. We see at the same time that different varieties (resp., quasivarieties) of groups correspond to different varieties (resp., quasivarieties) of quasigroups.

3) One can enrich the structure of a reversible automaton by supposing that the set of input signals B is a quasigroup, and that we are given a representation of this quasigroup on the set A of states. In addition, this representation must be assumed to be connected to the corresponding operation $*$.

THEOREM 12. Every reversible automaton (A, B, C) is isomorphic to some regular automaton (Q, Q, Q) , where Q is a loop.

Proof. Let (a, b, c) be an identity subautomaton of a reversible automaton (A, B, C) . If $x, y \in A$, then $y * b \in C$, $a \setminus (y * b) \in B$, $x * [a \setminus (y * b)] \in C$, $\{x * [a \setminus (y * b)]\} / b \in A$. So, by means of the elements $a \in A$ and $b \in B$, we can define a binary operation on the set A :

$$x \cdot y = \{x * [a \setminus (y * b)]\} / b.$$

In a similar manner, we see that the same pair defines the following binary operations $\circ$ and $\square$ on the sets B and C , respectively:

$$\begin{aligned} x \circ y &= a \setminus \{[a * x] / b\} * y, \\ x \square y &= (x / b) * (a \setminus y). \end{aligned}$$

It is easy to verify that $A(\cdot)$, $B(\circ)$, and $C(\square)$ are loops, with identify elements a , b , and c , respectively.

By using R_b and L_a ($R_b: a' \rightarrow a' * b$, $L_a: b' \rightarrow a * b'$ for all $a' \in A$ and $b' \in B$), we can rewrite as follows the equations that determine the operations $\cdot$, $\circ$, and $\square$:

$$\begin{aligned} (x \cdot y)^{R_b} &= x * y^{R_b L_a^{-1}}, \\ (x \circ y)^{L_a} &= x^{L_a R_b^{-1}} * y, \\ x \square y &= x^{R_b^{-1}} * y^{L_a^{-1}}. \end{aligned}$$

It follows from these last equations that the automata (A, A, A) , (B, B, B) , and (C, C, C) are isomorphic to the reversible automaton (A, B, C) , and the loops $A(\cdot)$, $B(\circ)$, and $C(\square)$ are isomorphic to each other:

$$\begin{aligned} (x \cdot y)^{R_b L_a^{-1}} &= x^{R_b L_a^{-1}} \circ y^{R_b L_a^{-1}}, \\ (x \cdot y)^{R_b} &= x^{R_b} \square y^{R_b} \quad \text{for all } x, y \in A. \end{aligned}$$

For $A = B = C = Q$, we obtain that the quasigroup $Q(*)$ is LP-isotopic to the loop $Q(\square)$.

Remark Inserted in Proof. The author has proved at this point that, over an arbitrary ring, the condition of a quasivariety being saturated relative to output signals is equivalent to its being defined by quasi-identities that do not involve any output variables. However, the hereditary property of being finitely based, contained in Theorems 5 and 6, may fail. As a consequence of this result, the problem mentioned on p.326 has been solved.

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STRONG LINEAR CONVEXITY.

I. DUALITY OF SPACES OF HOLOMORPHIC FUNCTIONS

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UDC 517.55

INTRODUCTION

1. Let $\mathcal{D}$ be an open set in $\overline{\mathbb{C}}$. The space $H(\mathcal{D})$ consists of functions which are holomorphic in $\mathcal{D}$, and convergence in $H(\mathcal{D})$ by definition means uniform convergence on each compact subset of $\mathcal{D}$.

Let $\mathcal{F}$ be an arbitrary set in $\overline{\mathbb{C}}$. We define the space $H(\mathcal{F})$ as the inductive limit of the spaces $H(\mathcal{D})$ over all open $\mathcal{D}$, containing $\mathcal{F}$. The space $H_0(\mathcal{F})$ is defined as the closed subspace of $H(\mathcal{F})$, consisting of functions which vanish at infinity if $\infty \in \mathcal{F} \subset \overline{\mathbb{C}}$, and is identified with $H(\mathcal{F})$ if $\mathcal{F} \subset \mathbb{C}$. By $H_0^*(\mathcal{F})$ we denote the space of continuous linear functionals on $H_0(\mathcal{F})$. For each $T \in H_0^*(\mathcal{F})$, the formula

$$\varphi(\xi) = T((\xi - z)^{-1}) \quad (1)$$

defines a function $\varphi \in H_0(\overline{\mathbb{C}} \setminus \mathcal{F})$, the indicatrix of this functional.

The basic result (cf. [1, 2]) of the theory of duality of spaces of holomorphic functions developed in [3-5] asserts that for a set $\mathcal{F}$ from a certain class which includes all open and all compact sets in $\overline{\mathbb{C}}$, the correspondence (1) establishes an isomorphism of spaces $H_0^*(\mathcal{F})$ and $H_0(\overline{\mathbb{C}} \setminus \mathcal{F})$. Examples of the use of this theorem to prove Runge's theorem, the theorems on interpolation and ideals, and the theorem of Poincaré-Aronszajn on the separation of singularities are given in [2, 6].

Considering multidimensional analogs of the duality theorem, we shall assume that $\mathcal{F}$ is holomorphically nonextendable, i.e., for each $z \notin \mathcal{F}$ there exists an $f \in H(\mathcal{F})$, which cannot be extended analytically to the point z . We also require that $\mathcal{F}$ be open or a compact set. These requirements on $\mathcal{F}$ will be understood in the formulations of all assertions of the paper.

2. A generalization of the correspondence (1) to functions of severable variables is defined in [7, p. 433] by the formula

$$\varphi(\xi) = T((\xi_1 - z_1)^{-1} \dots (\xi_n - z_n)^{-1}). \quad (2)$$

We find for which sets this generalization leads to a multidimensional duality theorem. For this we fix a set $\mathcal{F} \subset \mathbb{C}^n$ and we consider the collection $\mathcal{S}$ of points at which the indicatrices of all functionals from $H^*(\mathcal{F})$ are defined. For the correspondence (2) we consider the Dirac functionals (δ -functions) on $H(\mathcal{F})$, using which it is easy to see that

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